1. Introduction

Quick commerce focuses on ultra-fast delivery (10–15 minutes) using hyperlocal warehouses and real-time systems.

Key Concepts

1. Hyperlocal delivery model
2. Real-time inventory tracking
3. Customer-first experience
4. Low latency system design

2. Existing nopCommerce Architecture

Architecture Layers

1. Presentation Layer – UI and Controllers
2. Service Layer – Business logic
3. Data Layer – Database operations
4. Domain Layer – Core entities

Explanation

This layered architecture ensures separation of concerns but is not optimized for real-time and location-based operations.

3. Limitations of Current System

1. Global inventory without location awareness
2. No real-time stock updates
3. No delivery time estimation
4. Not optimized for quick commerce

4. Quick Commerce Requirements

1. 10–15 minute delivery capability
2. Real-time inventory validation
3. Location-based filtering

4. High scalability and performance

5. Proposed Enhancements

New Components

1. StoreLocation (latitude, longitude)
2. ProductInventory mapping
3. Geo-location service
4. Delivery ETA service
5. Caching (Redis)
6. Message Queue (Kafka/RabbitMQ)

6. Updated Architecture

Frontend captures user location and backend processes it to filter products and ensure accurate delivery.

Flow Highlights

1. Location-based product filtering
2. Real-time stock validation
3. Efficient order routing

7. Request Flow

1. User enters location
2. Controller receives request
3. Service filters nearest store
4. Inventory validated
5. Response returned with ETA

8. Database Changes

1. StoreLocation table
2. ProductInventory table
3. DeliveryZone table
4. Optimized indexing for performance

9. Scalability Improvements

1. Microservices architecture
2. Caching layer
3. Asynchronous processing
4. Load balancing

10. Performance Optimization

1. Reduced database calls
2. Optimized APIs
3. Efficient queries
4. Caching mechanisms

11. Benefits

1. Faster delivery
2. Better user experience
3. High scalability
4. Competitive advantage

12. Conclusion

Transforming nopCommerce into a quick commerce platform enables modern, scalable, and efficient delivery systems.